Omega MS3

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Circuit Purpose

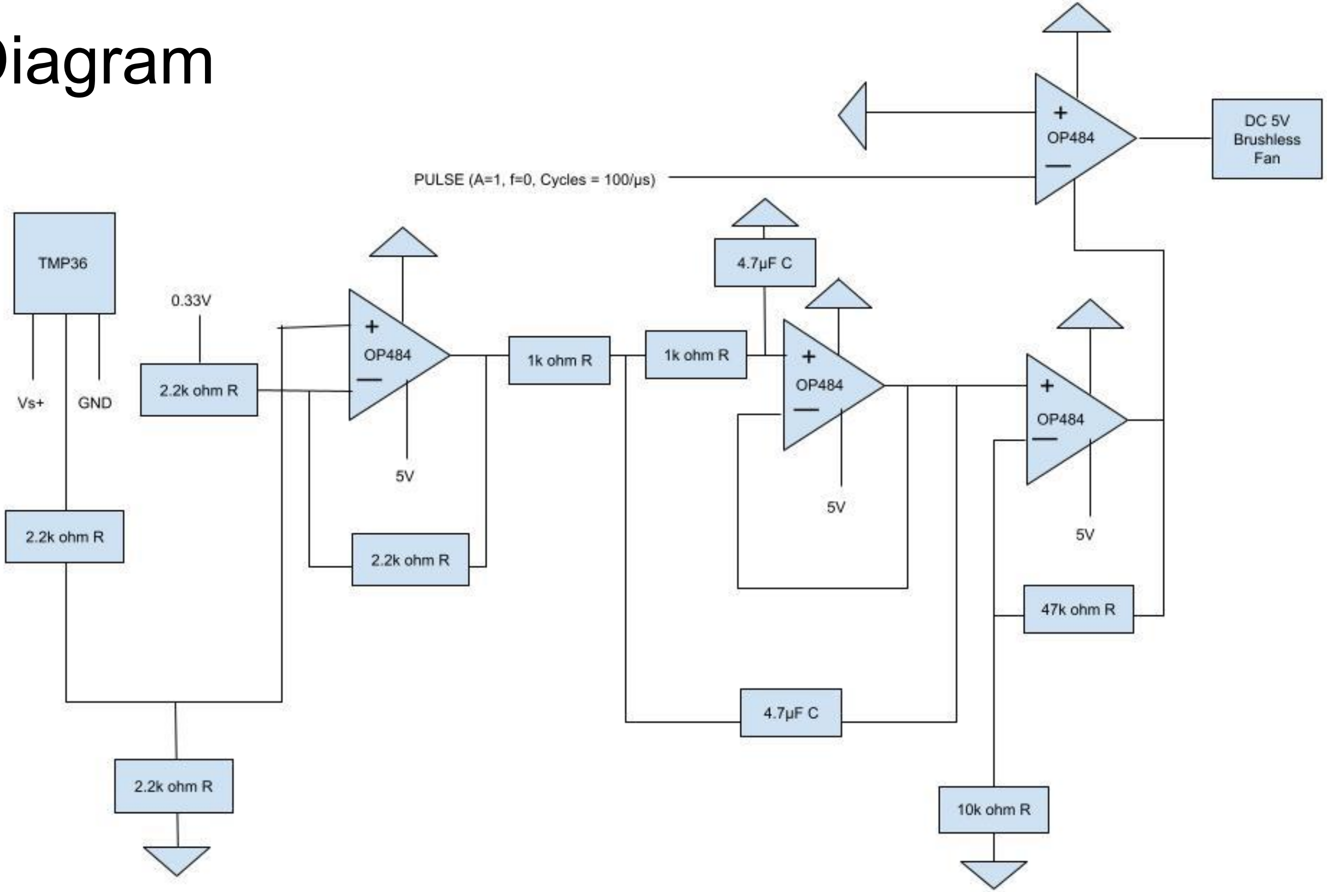
A TMP36 temperature sensor produces a voltage proportional to temperature, which is first conditioned using a OP484 differential amplifier. A threshold voltage is set, proportional to a specific temperature, so when the TMP36 voltage is below this point, the output is 0V, and when it is above, it is equal to V(TMP36)-V(threshold)

The signal then passes through a second order active low-pass filter to remove noise and prevent unstable behavior. After filtering, the voltage is then amplified by a non-inverting amplifier with a gain of ~5.7x

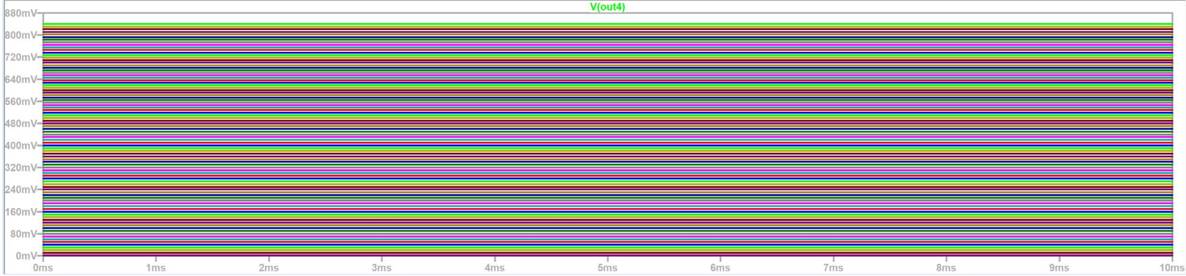
Once the voltage is amplified, it runs into a NPN transistor which allows us to increase the current. This is because the 5V motor requires 200mA while the op amps supply a low 20-40mA. This is then ran into a 5V DC motor.

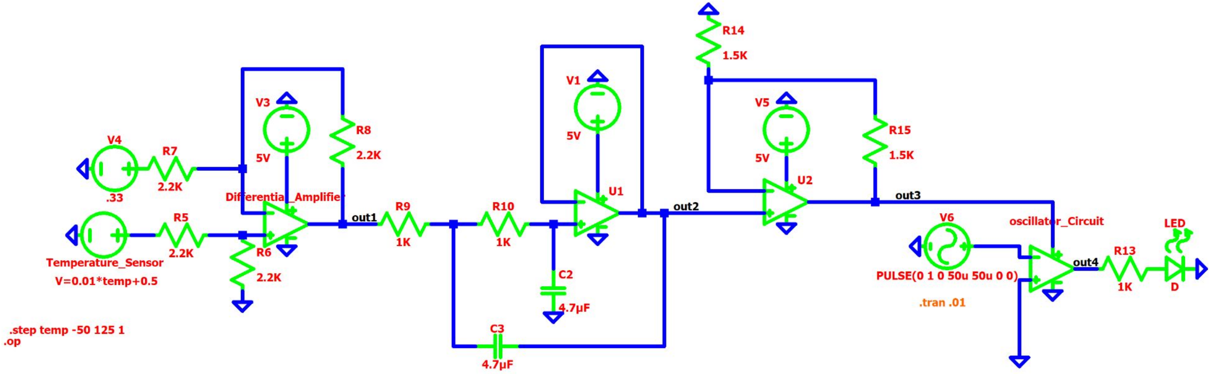
In the end the Fan will turn on after a threshold temperature is reached, and then scale its speed with temperature afterwards.

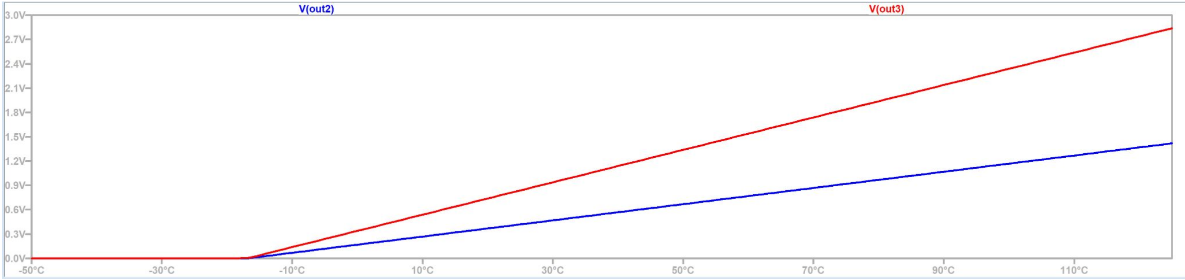
Block Diagram



Simulation


V(out4)
880mV
800mV
720mV
640mV
560mV
480mV
400mV
320mV
240mV
160mV
80mV
0mV
0ms
1ms
2ms
3ms
4ms
5ms
6ms
7ms
8ms
9ms
10ms


R14
1.5K
V1
5V
V3
5V
R8
2.2K
V5
5V
R15
1.5K
V4
.33
R7
2.2K
Differential_Amplifier
U1
U2
out1
R9
1K
R10
1K
out2
out3
R5
2.2K
V6
oscillator_Circuit
LED
Temperature_Sensor
V=0.01*temp+0.5
R6
2.2K
C2
4.7µF
PULSE(0 1 0 50u 50u 0 0)
out4
R13
1K
D
.tran .01
.step temp -50 125 1
.op
C3
4.7µF


V(out2)
V(out3)
3.0V
2.7V
2.4V
2.1V
1.8V
1.5V
1.2V
0.9V
0.6V
0.3V
0.0V
-50°C
-30°C
-10°C
10°C
30°C
50°C
70°C
90°C
110°C

Demonstration

Second-Order Active Filter

For the second order active filter, it doubles the roll-off rate from the first order passive filter we had is MS2. The roll-off rate goes from -20 dB per decade to -40 dB per decade. On top of this, the original block after the passive filter would draw a small amount of current from the passive filter, affecting the filter's operation. This would throw off the cutoff frequency. The new active filter will not do this. We used 1K ohm resistors and 4.7uF capacitors in this filter. This will create a cutoff frequency of ~33.86 Hz using the formula f=1/(2*pi*R*C).

Oscillator Circuit (Astable Multivibrator)

DC motors use electromagnetic coils that have an inherent inductance. On top of this the fan itself has physical inertia. This causes the fan to stall out when the voltage is low. To combat this, an oscillator circuit will give it the “push” it needs. The motor cannot turn on and off for every pulse, so it “averges out” these pulses to make it feel like a continuous 50% of what is being supplied. For this to work the frequency should be in the range of 20 to 25 kHz

Problems Encountered and Solutions

- We encountered a problem regarding the gain of our second-order active filter
    - We initially tried to amplify the voltage via the second-order active filter
    - However, when the gain of a second-order active filter is greater than 1, the output becomes unstable, and once it becomes greater than 3, the output becomes an oscillator circuit
    - We fixed this by making the gain of the filter 1, and adding a non-inverting amplifier after the filter with a gain of roughly 5.7
- We also encountered an issue when implementing the non-inverting amplifier
    - We initially tried to replicate the non-inverting amplifier we used in MS2 with the OP97
    - However, since the voltage that would be outputted would be less than 1V, the OP97 (non-rail-to-rail) would output the incorrect voltage
    - We fixed this by implementing the non-inverting amplifier in our OP484

Design Choices

- Oscillator Circuit(Astable Multivibrator)
  - Allows for better control and better torque for powering the 5V DC motor
- Second-Order Active Filter
  - We found that the first order passive filter from MS2 was still letting in some noise. To eliminate this we chose to incorporate a second-order active filter which allows us to better filter the output of the differential amplifier.
  - More accurate cutoff frequency
- 5V DC Brushless Fan
  - Works in the proper voltage range for our power supply (~3-5V)
  - The fan only draws 0.2A of current, meaning it won't overwhelm the circuit's power supply
  - The brushless fan has a longer operational life since there are no brushes to wear out, and less electrical noise that could interfere with the analog output of the TMP36 sensor
  - Simple to implement with a breadboard because of the power and ground wires